

Hemanth Kumar Mangalapurapu

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Summary

Full-Stack AI Software Engineer with 5 years building production platforms and distributed backend systems. Specializes in RAG architecture, multi-agent orchestration (LangGraph, MCP), and end-to-end delivery across Python/Go backends and React/Next.js frontends - from an enterprise LLM knowledge platform, to a distributed container orchestrator in Go, to a clinical AI system combining computer vision diagnostics. Comfortable owning a feature from model and backend design.

Technical Skills

Generative AI & LLM Engineering: RAG (hybrid retrieval, Reciprocal Rank Fusion, cross-encoder reranking), LangChain, LangGraph, Model Context Protocol, DSPy, Prompt Engineering, Azure OpenAI, Amazon Bedrock, Ollama

Machine Learning: PyTorch, TensorFlow, MONAI, CNN (DenseNet-121), NVIDIA Triton Inference Server, Matplotlib

Languages: Go (Golang), Python, C++, Java, TypeScript/JavaScript, React.js, Next.js, SQL

Backend & APIs: Node.js, Express, FastAPI, REST APIs, GraphQL, gRPC, Celery, WebSockets, NestJS

Databases & Vector Stores: PostgreSQL, DynamoDB, MongoDB, Redis, ChromaDB, Pinecone, pgvector

Cloud, Data & DevOps: AWS (Lambda, Fargate, ECS, EC2, S3, DynamoDB, AppSync, IAM, CloudWatch), Azure, Docker, CI/CD, Terraform, Grafana, Prometheus, Kafka, Databricks, Apache Spark

Experience

LeapGen AI | AI Python Engineer | Ashburn, VA

May 2025 - Present

- Built Retrieval-Augmented Generation pipelines using LangChain, vector databases, and document embedding/ingestion workflows, improving knowledge-retrieval accuracy and reducing irrelevant chatbot responses.
- Orchestrated AI workflow automation using LangGraph and Model Context Protocol (MCP), driving document analysis, multi-step reasoning, and task execution across business processes.
- Built production FastAPI services integrating OpenAI GPT-4o and Azure OpenAI into enterprise AI workflows, optimizing prompt orchestration, context management, and API performance for reliable LLM inference.
- Improved AI application observability by implementing structured logging, tracing, and performance analysis.
- Deployed and maintained Azure infrastructure using Docker, Terraform, and GitHub Actions, enabling CI/CD delivery.
- Delivered features across Seismic(healthcare platform - billing/subscription management, clinical workflows, system integrations) and DataCoffee(AI data-pipeline builder), owning implementation from backend APIs to production deployment.

SmartRevIQ | Software Engineer | Delaware, DE

Jun 2023 - Apr 2025

- Built scalable asynchronous data pipelines and REST APIs on AWS DynamoDB with client-side validation, processing thousands of enterprise pricing transactions at sub-3-second query performance for analytics workflows.
- Designed a dynamic form-rendering engine with nested validation and reusable schemas, powering new pricing UI flows across the platform and Built Dashboards with drill-down filtering, surfacing pricing analytics to identify margin leakage.
- Automated enterprise pricing workflows by deploying AI agents with LangGraph, reducing operational overhead.
- Implemented SSO authentication and Role-Based Access Control(RBAC), securing multi-tenant access to sensitive datasets.

Speeler Technologies | Software Engineer | Hyderabad, India

Apr 2021 – May 2023

- Architected and led the redesign of GraphQL backend services for a high-traffic e-commerce platform, implementing DataLoader-style resolver batching - AWS AppSync response caching to cut P95 latency by 30%.
- Designed a multi-tenant DynamoDB single-table architecture with sparse GSIs and composite sort keys, eliminating hot-partition bottlenecks and sustaining sub-10ms reads under high-cardinality access patterns, ensuring idempotency.
- Engineered a fault-tolerant, event-driven image-processing pipeline on AWS Lambda and SQS with dead-letter queue handling, cutting infrastructure costs by 70% while processing 2M+ monthly events at 99.99% data durability.
- Built a real-time collaborative Business Automation & Data Processing E-Commerce Platform with React and WebSockets for concurrent users, reducing perceived latency by 50ms.
- Migrated EC2 workloads to AWS ECS Fargate with Blue/Green deployments and container-level health checks wired into CI/CD, achieving zero-downtime releases across a 50K+ user production system.
- Built a TensorFlow-based demand-forecasting model for e-commerce inventory planning, visualizing forecast trends and accuracy with Matplotlib for stakeholder reporting.

Projects

HemGPT - Enterprise RAG Knowledge Base | Python, LangChain, ChromaDB, Pinecone, FastAPI, React GitHub

- Built a dual-mode Retrieval-Augmented Generation platform supporting both a fully local, privacy-first backend (Ollama, DSPy, ChromaDB) and a cloud backend (Pinecone), using a hybrid retrieval pipeline - dense vector search, BM25 sparse retrieval, and Reciprocal Rank Fusion re-ranking with a cross-encoder - to cut query response time.
- Built an async ingestion pipeline with FastAPI and Celery workers using per-document-type chunking and Chain-of-Thought to maximize context relevance, embedding a 10K-document corpus in under 10 minutes without blocking the UI.
- Deployed the inference layer with Server-Sent Events (SSE) streaming to eliminate perceived wait time for end users.

Deploy.sh - Distributed Container Orchestration Engine | Go, gRPC, PostgreSQL, Docker, Grafana GitHub

- Engineered a custom distributed control plane in Go with a high-frequency gRPC heartbeat and sliding-window health state machine, cutting node-failure detection latency from ~2s to under 30ms.
- Built a bin-packing scheduler (First-Fit Decreasing heuristic) that raised simulated hardware utilization from 50% to 80%, validated against 100+ container placement scenarios.
- Implemented a Write-Ahead Log in PostgreSQL with fsync-based durability, guaranteeing zero data loss across simulated controller-failover scenarios with recovery in under 5 seconds.
- Built a CLI and a Grafana/Prometheus dashboard exposing real-time cluster state and node metrics via RESTful API.

Sunx - AI-Powered Radiography Prediction & Clinical Decision Support | Python, PyTorch, FastAPI, Next.js GitHub

- Built and fine-tuned a DenseNet-121 vision model achieving ~80% classification accuracy and an AUC of 0.91 across 14 chest X-ray pathology classes, served via NVIDIA Triton Inference Server for downstream clinical reasoning.
- Engineered a Retrieval-Augmented Generation layer using LangChain and pgvector (1536-dimensional embeddings) to ground treatment recommendations in real clinical guidelines (AHA/ACC, IDSA/ATS) with source citations.
- Built a HIPAA-aware DICOM ingestion and ETL pipeline (pydicom, MONAI) with PHI anonymization, backed by an async Celery-Redis task queue to keep inference non-blocking.
- Developed a Next.js/TypeScript frontend with a Cornerstone.js supporting WebGL-based pan/zoom/window-leveling.

ShuttleNow - Real-Time Transit Booking App | MongoDB, Express.js, React, React Native, Node.js, Redis, Stripe GitHub

- Architected a fault-tolerant booking engine that cut seat double-booking conflicts from a 12% rate under burst load to 0.5%, using Redis Redlock distributed locking with lease expiration and Optimistic Concurrency Control.
- Delivered real-time seat-inventory updates to 1,000+ concurrent clients at under 50ms latency via a bidirectional Web-Socket layer instead of polling backed by MongoDB change-stream event diffing.
- Integrated Stripe payments with PCI-compliant webhook verification across both the React web app and a React Native mobile client, using serverless functions for asynchronous payment confirmation.

Education

Auburn University at Montgomery - Master of Science, Computer Science

Aug 2023 - May 2025

MLRIT - Bachelor of Technology, Computer Science & Engineering

Jul 2018 - Aug 2022